

CareFlow SQL Reporting Notes

Supporting artefact for OPD reporting and data analysis examples

Purpose: This artefact shows how CareFlow data can answer common OPD operational questions such as daily visit count, department queue load, pending lab payments, external report tracking, pharmacy dispensing gaps, and patient visit history.

Scope note: The queries are simplified examples for portfolio demonstration. Table and column names follow the CareFlow logical data model, and minor naming adjustments may be made in a real database implementation.

MySQL Function Notes

Function	Simple Meaning	Used For
CURRENT_DATE	Today's date only	Filtering today's visits or orders
CURRENT_TIMESTAMP	Current date and time	Calculating how long something has been pending
TIMESTAMPDIFF(MINUTE, start, end)	Difference between two times in minutes	Wait time or pending duration
LEFT JOIN	Keep the main record even if matching child data is missing	Finding missing external reports or undispensed items
CASE	Creates readable output based on conditions	Showing Not Dispensed / Partially Dispensed

Query 1 — Daily OPD Count by Visit Type

Business question: How many OPD visits happened today, grouped as New Visit or Follow-up?

```
SELECT
    VisitType,
    COUNT(VisitID) AS TotalVisits
FROM Visits
WHERE VisitDate = CURRENT_DATE
GROUP BY VisitType;
```

Plain-English Explanation

VisitType: The category of the visit: New Visit or Follow-up

COUNT(VisitID): Counts how many visit records are present

CURRENT_DATE: Filters the result to today only

GROUP BY VisitType: Shows separate counts for each visit type

Example Output

Column 1	Column 2
New Visit	35
Follow-up	112

Query 2 — Active Token Count by Department

Business question: Which department currently has the highest active OPD queue?

```
SELECT
    DepartmentCode,
    COUNT(VisitID) AS ActiveTokens
FROM Visits
WHERE VisitStatus = 'Active'
    AND VisitDate = CURRENT_DATE
GROUP BY DepartmentCode
ORDER BY ActiveTokens DESC;
```

Plain-English Explanation

VisitStatus = Active: Includes only patients still waiting or in progress

VisitDate = CURRENT_DATE: Includes today's queue only

GROUP BY DepartmentCode: Counts active tokens separately for each department

ORDER BY ActiveTokens DESC: Shows the busiest department first

Example Output

Column 1	Column 2
CARDIO	11
ORTHO	8
GENERAL	7

Query 3 — Pending Lab Payments

Business question: Which non-urgent in-house lab orders are still waiting for payment?

```
SELECT
    LabOrderID,
    VisitID,
    TestName,
    PaymentStatus,
    TIMESTAMPDIFF(MINUTE, OrderTimestamp, CURRENT_TIMESTAMP) AS MinutesPending
FROM LabOrders
WHERE LabType = 'InHouse'
    AND PaymentStatus = 'Pending'
    AND UrgencyFlag = 'N'
    AND DATE(OrderTimestamp) = CURRENT_DATE
ORDER BY OrderTimestamp;
```

Plain-English Explanation

LabType = InHouse: Only hospital lab orders need hospital payment tracking

PaymentStatus = Pending: Finds unpaid orders

UrgencyFlag = N: Excludes urgent cases because urgent care should not wait for payment

TIMESTAMPDIFF: Shows how long payment has been pending

Example Output

Column 1	Column 2	Column 3
LO-60041	CBC	28 mins
LO-60055	Lipid Profile	14 mins

Query 4 — External Reports Pending Upload

Business question: Which external lab orders have no uploaded report yet?

```
SELECT
    L0.LabOrderID,
    L0.VisitID,
    L0.TestName,
    L0.OrderTimestamp
FROM LabOrders L0
LEFT JOIN LabReports LR ON L0.LabOrderID = LR.LabOrderID
WHERE L0.LabType = 'External'
    AND LR.ReportID IS NULL
    AND DATE(L0.OrderTimestamp) = CURRENT_DATE
ORDER BY L0.OrderTimestamp;
```

Plain-English Explanation

LEFT JOIN: Keeps the lab order even when no report exists

LR.ReportID IS NULL: Finds orders without a matching uploaded report

LabType = External: Only external diagnostic orders are included

Example Output

Column 1	Column 2	Column 3
LO-60059	MRI Lumbar Spine	Pending Upload
LO-60063	Blood Culture	Pending Upload

Query 5 — Pharmacy Dispensing Gap

Business question: Which medicines were not dispensed or only partly dispensed today?

```
SELECT
    PLI.DrugName,
    PLI.Dosage,
    IFNULL(PDLI.QuantityDispensed, 0) AS QuantityDispensed,
    CASE
        WHEN PDLI.DispenseLineID IS NULL THEN 'Not Dispensed'
        WHEN PDLI.PartialFlag = 'Y' THEN 'Partially Dispensed'
        ELSE 'Dispensed'
    END AS DispenseStatus,
    PDLI.PartialReason
FROM PrescriptionLineItems PLI
JOIN ePrescriptions EP ON PLI.PrescriptionID = EP.PrescriptionID
LEFT JOIN PharmacyDispenseLineItems PDLI ON PLI.LineItemID = PDLI.LineItemID
WHERE DATE(EP.PrescriptionTimestamp) = CURRENT_DATE
AND (PDLI.DispenseLineID IS NULL OR PDLI.PartialFlag = 'Y');
```

Plain-English Explanation

IFNULL: Shows 0 when no dispense record exists

CASE: Converts database conditions into readable status text

LEFT JOIN: Keeps prescribed medicines even if pharmacy has not dispensed them

PartialFlag = Y: Finds partly supplied medicines

Example Output

Column 1	Column 2	Column 3
Metformin 500mg	0	Not Dispensed
Atenolol 25mg	15	Partially Dispensed

Query 6 — Patient Visit History Lookup

Business question: What previous visits, consultation summaries, and reports belong to one patient?

```
SELECT
    V.VisitID,
    V.VisitDate,
    V.DepartmentCode,
    CN.ChiefComplaint,
    CN.Assessment,
    CN.Plan,
    LR.ReportSource,
    LR.ReportDate
FROM Visits V
LEFT JOIN ConsultationNotes CN ON V.VisitID = CN.VisitID
LEFT JOIN LabOrders LO ON V.VisitID = LO.VisitID
LEFT JOIN LabReports LR ON LO.LabOrderID = LR.LabOrderID
```

```
WHERE V.PatientID = 'PT-10041'  
ORDER BY V.VisitDate DESC;
```

Plain-English Explanation

WHERE V.PatientID: Filters history to one patient

LEFT JOIN: Keeps visits even if notes or reports are missing

ReportSource: Shows whether a report is In-House, External, or Patient-Uploaded

ORDER BY VisitDate DESC: Shows most recent visit first

Example Output

Column 1	Column 2	Column 3	Column 4
V-20260715	General Medicine	Acute pharyngitis	InHouse
V-20260628	General Medicine	Follow-up	External

Portfolio Use Note

These queries are included to show reporting logic and data understanding from a Business Analyst perspective. They are not production-optimised SQL scripts. In a real project, the development/data team would adjust naming, indexing, performance, and database-specific syntax.